Transfer Learning for Buying a House

Is this house fairly priced? Reuse models already trained on millions of photos and texts

Transfer learning **House details, photos and text** **Uses Weeks 2–4, 7 and 9**

The task. A first-time buyer sees a house advertised for €340,000. Is that a fair price? You will predict a fair price from the listing's details, its photos and its written description. Instead of training from scratch, you will reuse models that were already trained on millions of other images and texts, and show when your prediction can be trusted.

THE DATA

In Week 9, the teaching team gives you recent house listings from one region. Each has details (size, bedrooms, location, energy rating), several photos, a written description, and the asking price you will predict. The data is not public, so no AI assistant has seen it. Photos can show people, car number plates and house numbers: blur them in any example you show.

WHAT YOU WILL DO

1. As a baseline, predict the price from the details alone, with linear regression and a tree-based model (Week 2).
2. Turn each photo and each description into a list of numbers, using pretrained image and text models (Week 9).
3. Adapt the pretrained models in three ways: train only a new final layer, fine-tune the whole model, and use LoRA. Compare their error, size, training time and energy use, and say which you would choose (Week 9).
4. Try one way of creating extra training photos, such as flipping or cropping them, and see whether it helps.

WHAT YOU CODE YOURSELF

The small network that turns the image and text numbers into a price: write its **backpropagation** and **optimiser step** yourself and check them against PyTorch. Use **PCA via SVD** on the photo numbers to see which houses the pretrained model thinks look alike, and check it against scikit-learn.

IS IT FAIR AND SAFE?

Is the model more wrong for cheaper areas? Measure the error as a percentage of the price for each neighbourhood and price band. Location can stand in for things like ethnicity or income, so check for this, try one fix and measure what it costs. Use SHAP to show which details move the price. Write a note on the rules for collecting listings, privacy in photos, and why this is not financial advice.

QUESTIONS TO PREPARE FOR

Why did you choose that way of adapting the model, and what did each option cost? What does the image model notice that the listing details miss? Who could be hurt if buyers trusted your price?

COMMON MISTAKES

The same house listed twice, once in training and once in testing. The price written inside the description. Choosing an adaptation method on accuracy alone, without counting its cost.

WEEK-BY-WEEK PLAN

Week 9: baseline trained; photo and text numbers extracted; PCA tested. **Week 10:** the three adaptation methods compared, with their costs side by side. **Week 11:** extra-photos experiment; fairness check across areas with one fix; SHAP; model card and governance note. **By 6 January 2027:** report written; repository checked and submitted. **Week of 11 January:** 10-minute in-person interview.

TO AIM HIGHER

Give a price range instead of one number, and check how often the true price falls inside it. Train on older listings and test on newer ones, to see how fast the model goes out of date. Show which parts of a photo change the price. Estimate the energy each method used.